

The Complete Treatise on the Architecture of an Orderly Global Monetary Transition

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Abstract

An orderly global monetary transition is not principally a problem of printing a new currency. It is a systems problem involving representations of value, interest-rate markets, derivatives, claims, banking institutions, logistics, governance, and the physical economy. This treatise develops a compact architecture for such a transition. The central principle is that a monetary representation must not be confused with the real resources, institutional capacities, contracts, and logistics that give the representation economic meaning. The framework combines the generalized-claims queueing model, extremum-normalized stochastic-process representation, intrinsic-value and speculative-decision framework, quadratic transformations, the Ghoshian banking triad, and an entropic optimal-transport model of global port networks. The resulting architecture emphasizes stabilization before substitution, routing before abandonment, and orderly retirement rather than destructive discontinuity.

The treatise ends with “The End”

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1 The Central Problem

A reserve currency is not merely paper, an electronic balance, or a symbol. It is a representation embedded in a network of contracts, markets, institutions, payment systems, expectations, and real economic activity. The distinction between substrate and claim is therefore fundamental.

$$V_{\text{note}} \neq V_{\text{substrate}} \quad (1)$$

More generally, the value of a monetary instrument should be separated from the physical material on which it is recorded and from the institutional system that supports its use. The analysis of the dollar in the supplied monetary paper makes precisely this distinction: nominal representation is not identical to underlying economic reality [1].

The transition problem can therefore be stated as follows:

$$\text{Monetary transition} \neq \text{replacement of one physical object by another physical object.} \quad (2)$$

The objective is continuity of valuable functions while obsolete or negative-value structures contract, transform, or disappear.

2 Representation, Reality, and the Economic Stack

The monetary system can be represented as a layered architecture. A useful ordering is

$$\begin{aligned} \text{Real resources} &\longrightarrow \text{Logistics} \longrightarrow \text{Institutions,} \\ \text{Institutions} &\longrightarrow \text{Interest rates} \longrightarrow \text{Derivatives,} \\ \text{Derivatives} &\longrightarrow \text{Claims} \longrightarrow \text{Currencies.} \end{aligned} \quad (3)$$

This ordering is not a claim that every economy literally operates in this sequence. It is a dependency graph. A currency ultimately represents claims on an economic system whose physical and institutional capacities exist at deeper layers.

The monetary paper therefore provides a general methodological rule:

$$\text{Representation} \neq \text{reality represented.} \quad (4)$$

The practical implication is that a monetary transition should preserve the underlying functions before changing the representation through which those functions are recorded [1].

3 Why Interest Rates Come First

The first financial layer requiring stabilization is the interest-rate system. Interest rates determine discounting, valuation, funding costs, duration, convexity, and the pricing of a large family of derivatives.

A generic discount factor can be represented by

$$D(t, T) = \exp \left(- \int_t^T r(u) du \right). \quad (5)$$

For a zero-coupon claim with maturity value F , the corresponding continuously compounded value is

$$P(t, T) = FD(t, T). \quad (6)$$

A discontinuous monetary transition that changes the unit of account while the discounting and collateral system remains unstable risks creating valuation discontinuities even when the underlying real assets have not changed.

The stabilization sequence is therefore naturally represented as

$$\begin{aligned} &\text{Policy rates} \longrightarrow \text{Yield curves} \longrightarrow \text{Discount factors}, \\ &\text{Discount factors} \longrightarrow \text{Derivatives} \longrightarrow \text{Collateral and margin}, \\ &\text{Collateral and margin} \longrightarrow \text{Sovereign spreads} \longrightarrow \text{Currency transition}. \end{aligned} \quad (7)$$

This is an architectural ordering rather than a prescription for any particular central bank.

4 Interest-Rate Derivatives and Nonlinear Exposure

A derivative system introduces exposures that are nonlinear in the underlying rates. Duration is a first-order sensitivity, while convexity is a second-order sensitivity.

$$\Delta P \approx -DP \Delta y, \quad (8)$$

$$\Delta P \approx -DP \Delta y + \frac{1}{2}CP(\Delta y)^2. \quad (9)$$

Consequently, a transition framework must consider not merely the level of interest rates but also the geometry of the term structure and the nonlinear exposures of derivative books.

The same principle extends to swaps, caps, floors, swaptions, and other contingent claims. A monetary transition that ignores derivative exposures may transfer rather than eliminate systemic risk.

5 Claims: Positive, Zero, and Negative Value

The generalized-claims framework treats institutions as stochastic routing networks through which heterogeneous claims move [2]. A claim can have positive, zero, or negative economic value.

$$V(q) > 0 \quad \text{means that claim } q \text{ has positive net economic value.} \quad (10)$$

$$V(q) = 0 \quad \text{means that claim } q \text{ is economically neutral at the relevant valuation boundary.} \quad (11)$$

$$V(q) < 0 \quad \text{means that accepting or carrying claim } q \text{ imposes a net burden.} \quad (12)$$

The supplied intrinsic-value framework distinguishes intrinsic value from the market price of an asset [3]. Its speculative-decision extension further distinguishes undervaluation, overvaluation, zero intrinsic value, and negative intrinsic value [4]. These distinctions matter because a monetary transition should not treat every outstanding nominal claim as if it had identical economic substance.

The key architectural identity is

$$\text{Currency transition} \neq \text{claim resolution.} \quad (13)$$

Claims may instead be honored, repriced, netted, collateralized, restructured, transferred, written down, or extinguished according to their economic and contractual status.

6 Institutional Queues and Congestion

Let $R_t(\kappa)$ denote the backlog of claims at institutional channel κ . Let $A_t(\kappa)$ denote incoming claims and $\mu_t(\kappa)$ the service rate. The queue evolves according to

$$R_{t+1}(\kappa) = R_t(\kappa) + A_t(\kappa) - \mu_t(\kappa). \quad (14)$$

Capacity may be constrained by institutional quality and available capacity:

$$\mu_t(\kappa) \leq \theta_t(\kappa)\bar{C}(\kappa). \quad (15)$$

The supplied generalized-claims paper models routing as an endogenous choice influenced by backlog and channel characteristics [2]. A generic routing rule is

$$\kappa^*(q) \in \arg \max_{\kappa} U(q, \kappa; R_t(\kappa)). \quad (16)$$

Congestion feeds back into routing:

$$\frac{\partial U}{\partial R} < 0. \quad (17)$$

This produces a critical systems insight. If negative-value claims are simultaneously exported by many institutions, the receiving institutions can become congested. A transition can therefore fail even when each individual transfer appears rational.

A stylized stability condition is

$$\lambda < \lambda_c \implies \text{stable aggregate backlog,} \quad (18)$$

whereas

$$\lambda > \lambda_c \implies \text{potentially divergent backlog.} \quad (19)$$

The transition problem is consequently a network problem, not merely a balance-sheet problem.

7 Quadratics as Controlled Transformation Operators

Quadratic equations provide a compact algebraic mechanism for transforming two-root structures while retaining explicit control over invariants and ordering.

Consider

$$ax^2 + bx + c = 0, \quad a \neq 0, \quad (20)$$

with two distinct positive roots $\alpha > \beta > 0$. Define

$$S = -\frac{b}{a}, \quad (21)$$

$$\delta = \sqrt{\frac{b^2 - 4ac}{a^2}}. \quad (22)$$

Then

$$\alpha = \frac{S + \delta}{2}, \quad (23)$$

$$\beta = \frac{S - \delta}{2}. \quad (24)$$

Suppose the roots are transformed to

$$r_1 = \alpha + \mu, \quad (25)$$

$$r_2 = \beta + \nu. \quad (26)$$

The derived monic quadratic is

$$x^2 + \left(\frac{b}{a} - \mu - \nu\right)x + \left(\frac{c}{a} + \mu\nu - \frac{b(\mu + \nu)}{2a} + \frac{\nu - \mu}{2}\sqrt{\frac{b^2 - 4ac}{a^2}}\right) = 0. \quad (27)$$

The supplied quadratic paper shows that the symmetric component of the transformation shifts both roots, whereas the antisymmetric component changes their separation [5]. If

$$m = \frac{\mu + \nu}{2}, \quad (28)$$

$$d = \frac{\mu - \nu}{2}, \quad (29)$$

then m represents common translation while d represents differential displacement.

This is useful architecturally because controlled transformation can be performed at the coefficient level without requiring an uncontrolled reparameterization of the entire system.

8 Extremum-Normalized Stochastic Processes

The supplied extremum-normalized stochastic-process framework separates the geometry of a realized trajectory from its normalized shape [6]. For a continuous process S_t on the unit interval, define

$$H = \max_{0 \leq t \leq 1} S_t, \quad (30)$$

$$L = \min_{0 \leq t \leq 1} S_t. \quad (31)$$

Let τ_H and τ_L denote corresponding extremum times. The normalized process is

$$Z_t = \frac{S_t - t}{H - \tau_H}. \quad (32)$$

The reconstruction identity is

$$S_t = t + (H - \tau_H)Z_t. \quad (33)$$

The framework also defines the corresponding lower normalization and establishes a rank-one relation conditional on extremum geometry. Its crucial empirical warning is that the extrema are generally ex post quantities. Therefore, an ex post normalized path must not automatically be treated as an observable real-time trading signal [6].

This distinction is directly relevant to monetary transition analysis. Historical trajectories may be normalized and compared after realization, while forecasting and policy operations must respect the information available at the decision time.

The law of the normalized path can be represented as a mixture over extremum geometry:

$$P_Z(dz) = \int P_Z(dz \mid G = g)P_G(dg), \quad (34)$$

where

$$G = (H, L, \tau_H, \tau_L). \quad (35)$$

Thus uncertainty may be decomposed into uncertainty about global trajectory geometry and uncertainty about conditional path shape.

9 The Banking Architecture

The supplied Ghoshian banking-triad paper proposes three institutional components: a Gold Bank, a Reserve Bank, and a Central Bank [7]. In its notation, the three components satisfy the following structural relations:

$$\frac{dG}{dt} \geq 0, \quad (36)$$

$$\frac{dP}{dt} \geq 0, \quad (37)$$

$$\frac{dE}{dt} = 0, \quad (38)$$

where G denotes reserves, P denotes monetary momentum, and E denotes monetary energy within the proposed framework.

The paper interprets the complete triad as an institutional source of gravitational pull and argues that incomplete institutional structures can experience pressure toward convergence [7]. Whatever the empirical status of the proposed physical analogy, its architectural contribution is clear: monetary credibility depends on institutional structure rather than on the printed substrate of a currency.

A stylized convergence function in the supplied framework is

$$F_{\text{convergence}} = \kappa \frac{\text{Stability}_A - \text{Stability}_B}{\text{Distance}_{\text{institutional}}^2}. \quad (39)$$

This should be interpreted as a model within the supplied theory, not as an established empirical law.

10 Many Ports Rather Than One Harbor

A monetary transition ultimately operates on a physical economy. Global maritime logistics is therefore part of the architecture rather than an external detail. The supplied spherical-port-network model represents ports as vertices of a directed weighted graph and routes flows using logistic choice and entropy-regularized optimal transport [8].

Let f_{ij} denote flow from port i to port j , and let d_{ij} denote the corresponding great-circle distance. A simple logistic-chain cost is represented by

$$L_{ij} = f_{ij} d_{ij}. \quad (40)$$

A softmax routing probability can be written as

$$p_{ij} = \frac{\exp(-\lambda C_{ij})}{\sum_k \exp(-\lambda C_{ik})}. \quad (41)$$

The parameter λ controls sensitivity to routing cost. The supplied paper shows the limiting interpretation that small λ produces relatively diffuse routing, while large λ approaches cost-minimizing routing [8].

The equilibrium problem can be represented abstractly as

$$\min_{f \in \mathcal{F}} \left\{ \sum_{i,j} f_{ij} C_{ij} + \varepsilon \sum_{i,j} f_{ij} \log f_{ij} \right\}, \quad (42)$$

where $\mathcal{F}$ is the feasible transport set and the entropy term regularizes the routing problem. The corresponding Sinkhorn scaling provides an algorithmic route to equilibrium under the assumptions stated in the supplied paper [8].

The architectural lesson is straightforward: there are many ports, many routes, and many feasible channels. A transition should therefore not depend on a single harbor or a single institutional gateway.

11 Boston Harbor and the Difference Between Protest and Transition

The Boston Tea Party illustrates a general distinction between disrupting a system and constructing its successor. In December 1773, protesters in Boston Harbor destroyed tea belonging to the East India Company in opposition to British taxation and the political circumstances surrounding representation.

The historical episode is useful here not because it supplies a monetary-transition model, but because it illustrates a systems principle:

$$\text{Disruption} \neq \text{successful transition.} \quad (43)$$

Throwing cargo into a harbor can destroy an existing flow. It does not by itself provide another ship, port, insurance contract, payment system, settlement mechanism, or supply chain.

A constructive transition therefore requires the following sequence:

$$\text{Stabilize} \longrightarrow \text{Route} \longrightarrow \text{Transfer} \longrightarrow \text{Retire.} \quad (44)$$

The final operation is retirement rather than destruction wherever continuity of valuable functions can be preserved.

12 Governance, Legitimacy, and Collective Choice

The supplied treatise on quadratic voting describes a mechanism in which voters can purchase voting intensity at a quadratic cost [9]. If v_{ij} denotes the number of votes allocated by voter i to alternative j , a stylized cost is

$$C_{ij} = v_{ij}^2. \quad (45)$$

The mechanism is relevant here as an example of institutional design rather than as a universal solution to monetary governance. The paper emphasizes implementation constraints including credit allocation, fairness, technical infrastructure, strategic manipulation, collusion, and legitimacy [9].

The broader distinction is

$$\text{Mathematical feasibility} \neq \text{Institutional feasibility} \neq \text{Political feasibility.} \quad (46)$$

A technically coherent monetary architecture can therefore fail if its institutions cannot implement it or if the relevant populations and institutions do not accept its governing rules.

13 Welfare Economics and the Social Objective

A transition should not be evaluated solely by the survival of financial institutions. Its relevant objective is preservation of economically valuable activity and limitation of avoidable welfare loss.

Let W denote aggregate welfare and L denote transition loss. A generic objective is

$$\max W - L. \quad (47)$$

A more explicit decomposition is

$$L = L_{\text{settlement}} + L_{\text{liquidity}} + L_{\text{valuation}} + L_{\text{congestion}} + L_{\text{production}} + L_{\text{coordination}}. \quad (48)$$

This decomposition is conceptual. The supplied generalized-claims framework provides the queueing and congestion component, while the financial and logistics frameworks provide corresponding representations of valuation and routing risks [2,8].

14 Political Economy, Diplomacy, and Geopolitics

A reserve currency operates simultaneously as an economic instrument and as part of an international institutional network. Monetary transition therefore involves creditors, debtors, central banks, commercial banks, exchanges, clearing systems, governments, firms, households, and international institutions.

Diplomacy is particularly relevant to continuity because the transition requires mutual recognition of contracts, settlement conventions, and institutional jurisdictions. Political authority cannot be reduced to a mathematical variable, but political decisions can alter the feasible set of economic arrangements.

A useful systems representation is

$$\mathcal{F}_{\text{transition}} = \mathcal{F}_{\text{financial}} \cap \mathcal{F}_{\text{institutional}} \cap \mathcal{F}_{\text{legal}} \cap \mathcal{F}_{\text{logistical}} \cap \mathcal{F}_{\text{political}}. \quad (49)$$

The intersection formulation emphasizes constraint compatibility rather than political preference.

15 Warfare Economics and Strategic Resilience

A monetary system also exists under the possibility of geopolitical conflict. War can damage ports, energy systems, communications, financial infrastructure, shipping routes, and production capacity. Conversely, economic disruption can affect strategic capacity.

The relevant principle is resilience through redundancy. If a system depends on one route, one settlement mechanism, one currency, one port, or one institutional gateway, the failure of that component can propagate through the network.

A simple reliability representation is

$$R_{\text{system}} = 1 - P(\text{critical network failure}). \quad (50)$$

Adding independent or partially independent channels can increase resilience, although correlated failures can limit the benefit of redundancy. The point is architectural rather than military: continuity should not depend on a single point of failure.

16 Physics, Chemistry, Biology, and Psychology as Analogies

Interdisciplinary analogies are useful when they identify structural similarities without being mistaken for empirical identity.

In physics, conservation laws distinguish quantities that are transformed from quantities that are destroyed. In chemistry, reaction pathways describe transformations subject to conservation constraints. In biology, ecological networks exhibit flows, bottlenecks, adaptation, redundancy, and cascading failure. In psychology and psychiatry, institutions can be studied at the level of collective information processing and organizational behavior without making unsupported claims about the mental state of individual decision-makers.

The common abstraction is

$$\text{State}_{t+1} = \mathcal{T}(\text{State}_t, \text{flows, constraints, information}). \quad (51)$$

The analogy should not be interpreted as evidence that an economy literally obeys a physical conservation law. It is a modeling device for identifying transformation, constraint, and feedback.

17 Computer Science, Data Science, and Artificial Intelligence

The architecture is naturally represented as a directed dependency graph. Nodes represent institutional or economic states; edges represent transformations or flows.

$$G = (V, E). \quad (52)$$

A transition algorithm can therefore be described without assuming that the entire global economy is controlled by a single optimizer.

INPUT:

interest-rate state
derivative exposures
institutional queues
claims and valuations
banking capacities
logistics network
legal constraints
political constraints

1. Stabilize observable interest-rate infrastructure.
2. Map derivative sensitivities and collateral dependencies.
3. Classify claims by economic and contractual status.
4. Identify congested institutional channels.
5. Route claims toward feasible channels.
6. Preserve positive-value production and settlement flows.
7. Provide redundant payment and logistics channels.
8. Establish successor accounting and settlement conventions.
9. Transfer contracts under explicit conversion rules.
10. Retire obsolete structures gradually.
11. Monitor feedback, congestion, and failure propagation.
12. Update the transition state as new information arrives.

OUTPUT:

an orderly transition with minimized avoidable disruption.

The machine-learning interpretation is also natural. Softmax routing corresponds to probabilistic channel selection, while entropy regularization corresponds to controlled diversification of feasible routes [8]. The extremum-normalized framework provides a functional representation of realized trajectories [6]. The quadratic transformation provides an exact structure-preserving algebraic layer [5].

A generic architecture can be summarized as

$$\text{World} \longrightarrow \text{Data} \longrightarrow \text{Representation} \longrightarrow \text{Model} \longrightarrow \text{Decision} \longrightarrow \text{Institutional action.} \quad (53)$$

At every stage, representation error and information leakage must be controlled.

18 A Formal Stability Principle

The combined framework suggests a general proposition.

Proposition. An orderly monetary transition requires continuity of critical real and financial flows to the extent necessary for the system to remain within its feasible operating region.

Proof. Let $\mathcal{F}_t$ denote the feasible state set at time t . Suppose a transition produces a state outside $\mathcal{F}_{t+1}$. Then at least one required constraint is violated. Such a violation can manifest as settlement failure, insufficient liquidity, institutional congestion, contract discontinuity, or physical supply disruption. Hence a transition rule that does not preserve the relevant feasibility constraints cannot guarantee orderly continuation. Therefore an orderly transition requires preservation of the constraints defining the critical operating region. $\square$

The proposition is deliberately general. It does not select a particular currency, institution, government, or policy regime.

19 A Second Proposition: Claims Must Be Distinguished From Their Unit of Account

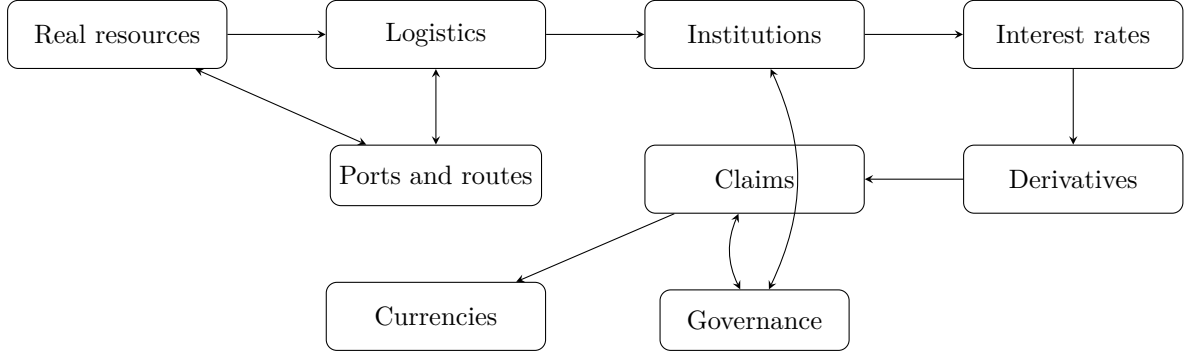
Proposition. Changing the unit in which a claim is denominated does not by itself determine the economic value of the underlying claim.

Proof. Let a claim be represented by a contractual payoff and let its accounting representation be changed through a mapping Φ . If the underlying contractual rights, obligations, collateral, counterparties, and settlement conditions remain invariant under the mapping, then the numerical representation

changes without necessarily changing the underlying economic object. Therefore nominal conversion and economic transformation are logically distinct operations. $\square$

This proposition formalizes the central distinction between representation and reality developed in the supplied monetary paper [1].

20 A Systems Diagram



The diagram emphasizes that monetary representation is embedded in a broader network. A currency transition is therefore a state transition of the network rather than a simple substitution of one label for another.

21 The Dignity Principle

The phrase “letting a reserve currency pass on with dignity” can be formalized as an orderly-retirement principle. The objective is not to preserve every nominal structure forever. Nor is the objective to destroy an existing structure merely because it has become obsolete.

The relevant distinction is

$$\text{Retirement of obsolete representation} \neq \text{destruction of valuable underlying relationships.} \quad (54)$$

An orderly transition should therefore seek to preserve productive capacity, contractual continuity, settlement capacity, logistics, institutional memory, and legitimate governance while allowing obsolete representations to contract or disappear.

22 What the Architecture Produces

The proposed architecture does not produce “more currency” as its principal output. Its output is a functioning network capable of carrying real economic activity through a change in monetary representation.

Table 1: Transition layers and primary objective

Layer	Primary architectural objective
Interest rates	Stabilize discounting and funding relationships.
Derivatives	Control nonlinear exposures, collateral, and margin dependencies.
Claims	Distinguish valuable, neutral, and burdensome obligations.
Institutions	Prevent congestion and preserve processing capacity.
Banking	Maintain credible reserve, liquidity, and policy functions.
Logistics	Preserve physical movement of goods and resources.
Ports	Maintain alternative routes and reduce single points of failure.
Governance	Establish legitimate rules for conversion and settlement.
Currencies	Provide stable units of account and settlement.

The resulting architecture can be summarized by the following objective:

$$\min (L_{\text{settlement}} + L_{\text{liquidity}} + L_{\text{congestion}} + L_{\text{valuation}} + L_{\text{production}} + L_{\text{coordination}}) \quad (55)$$

subject to continuity of the critical constraints of the economic system.

23 Conclusion

The central lesson is that a reserve-currency transition cannot be reduced to the question of which currency replaces which other currency. The deeper problem is how to preserve the real, financial, institutional, logistical, and political functions represented by the existing monetary architecture while allowing obsolete representations and burdensome claims to be transformed or retired.

The supplied theories fit together as different views of this problem. The monetary analysis separates substrate, representation, and institutional value [1]. The generalized-claims framework supplies stochastic routing and congestion [2]. The intrinsic-value and speculation papers distinguish economic value from market price [3,4]. The quadratic framework provides controlled transformation of structured quantities [5]. Extremum normalization separates realized path geometry from normalized shape while warning against look-ahead bias [6]. The banking-triad framework emphasizes institutional completeness [7]. The spherical-port model supplies a many-route logistics architecture based on entropy-regularized transport [8]. Quadratic voting provides an example of institutional mechanism design and highlights implementation and legitimacy constraints [9].

Together they imply a simple systems principle:

Preserve valuable structure; transform unstable structure;
 retire obsolete representation.

(56)

The desired output is consequently not another pile of nominal currency. It is a resilient monetary and economic operating architecture in which claims can be routed, institutions can process them, interest-rate markets can price them, logistics can move the underlying resources, and governance can maintain continuity while the monetary representation changes.

A Research Program

The framework suggests several empirical programs.

1. Estimate interest-rate transition risk from the term structure and derivative sensitivities.
2. Construct claim-level measures of positive, zero, and negative economic value.
3. Estimate institutional queue capacities and congestion thresholds.
4. Test whether routing diversification reduces systemic bottlenecks.
5. Estimate extremum geometry and normalized path distributions using leakage-free procedures.
6. Compare competing banking architectures using explicit institutional variables.
7. Estimate port-network resilience under route and capacity shocks.
8. Study the interaction between monetary conversion rules and contractual continuity.
9. Build simulation environments combining financial, institutional, and logistics networks.
10. Evaluate transition policies using welfare loss rather than nominal monetary volume alone.

These are research directions, not claims that the supplied models have already been empirically validated.

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Glossary

Claim

A contractual or institutional object representing a right, obligation, or contingent economic relationship.

Currency

A monetary representation used for accounting, settlement, exchange, or stores of value.

Derivative

A financial contract whose value depends on an underlying variable or set of variables.

Discount factor

A factor converting a future payoff into present value under a specified discounting convention.

Entropy regularization

A mathematical device that penalizes overly concentrated transport allocations and produces smooth routing solutions.

Extremum geometry

The extrema and their occurrence times in the normalized-process framework.

Institutional congestion

A state in which incoming claims approach or exceed an institution's processing capacity.

Liquidity

The capacity to meet payments or trade obligations without disproportionate disruption or price impact.

Monetary representation

The accounting or settlement form through which economic claims are expressed.

Negative-value claim

A claim whose carrying or acceptance imposes a net economic burden under the relevant valuation framework.

Numeraire

A unit relative to which economic values are expressed.

Reserve currency

A currency held by institutions as part of their international monetary and reserve arrangements.

Settlement

The completion of a financial or contractual exchange through the transfer or discharge of the relevant obligations.

Sinkhorn scaling

An iterative matrix-scaling procedure associated with entropy-regularized optimal transport.

Systemic risk

The possibility that disturbances propagate through interconnected institutions or markets and impair system-wide functioning.

The End